AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 22 · STUDENT LESSON

Train a Regression Model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Declare the candidates. Fit on training. Freeze before scoring.

Select and interpret a limited regression line while preserving evaluation and unit boundaries.

Keep one habit: Complete T/Vsource roles → candidate SSE search → pictured fixed prediction → baseline comparison → independent fresh line.

MAIN LESSON · SECTION 1

Which candidate line did the training evidence select?

> Choosing from a declared small set differs from fitting every possible line.

Learning goals

I can fit a line from a declared **training candidate set**.

I can separate **training** loss from validation comparison.

I can interpret **slope** and **intercept** with units.

Retrieve squared-error calculation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.
- How is MSE calculated?

End goal

By the end, I can explain train a regression model using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.
</details>



LEARNING TARGET **Training** points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Declare the candidate slopes and intercepts before scoring

> Select using training only, then apply the stored line unchanged.

Limited classroom fitting method and fictional scope

A line predicts y as m*x+b. In this classroom search we compare a small declared set of **slope**s and **intercept**s; this is not a full least-squares solver over all possible lines.

Use **training** error to choose a candidate, then freeze it before validation.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building **process**; this week examines **fitting** a line.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use train a regression model to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Use a complete training and validation fixture

> All observations needed for the lesson's promised calculations are supplied.

Complete source-role table

Role / ID	x in days	Observed height y in cm	Use
Training T1	0	1	Fit among declared A/B/C candidates
Training T2	1	3	Fit among declared A/B/C candidates
Training T3	2	5	Fit among declared A/B/C candidates
Validation V1	3	6	Score the frozen line and baseline
Validation V2	4	10	Score the frozen line and baseline

Candidate calculations and original search script

Training pairs are (0, 1),(1, 3),(2, 5). Candidate A has m 1 b 1, B m 2 b 0, C m 2 b 1. **Prediction**s are A [1, 2, 3], B [0, 2, 4], C [1, 3, 5]. **Training** squared-error sums are 5, 3, 0 respectively, so choose C.

Ties would use the earlier listed candidate.

```
training = [(0, 1), (1, 3), (2, 5)]
candidates = [(1, 1), (2, 0), (2, 1)]
def sse(line):
    m, b = line
    return sum((y - (m * x + b)) ** 2 for x, y in training)
chosen = min(candidates, key=sse)
print([sse(line) for line in candidates])
```

```
print(chosen)
# [5, 3, 0], then (2, 1); first candidate wins ties.
```

Trace assumptions,units and validation givens

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

The original W2 and W3 questions need validation observations as well as x values. The complete fixture now supplies V1=(3,6) and V2=(4,10); keep these separate from training. A:y=x+1, B:y=2x, C:y=2x+1 have training SSE 5,3,0 cm², so C wins. The script assumes finite numeric pairs, a nonempty candidate list and matching units.

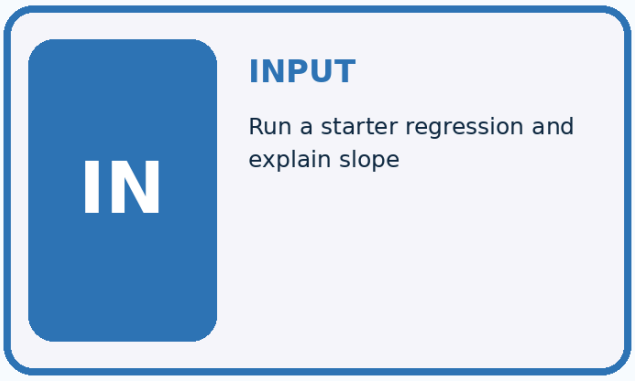
Its SSE is a sum, not MSE; dividing each training SSE by the same count would leave this ranking unchanged.

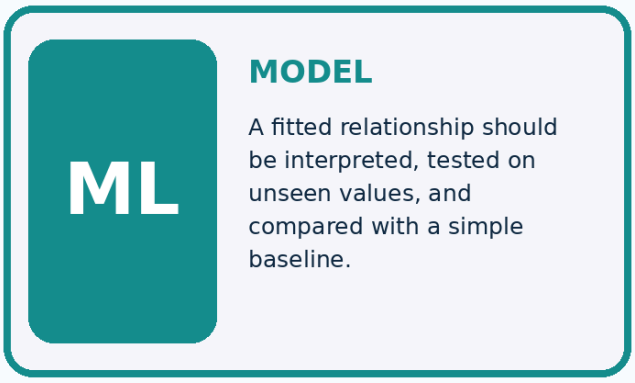
▼ Earlier wording, alternatives and picture transcripts

ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Run a starter regression and explain **slope**, **prediction**, residual, and MAE.

MODEL IDEA A fitted relationship should be interpreted, tested on unseen values, and compared with a simple baseline.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Model predicts height = 4 + 2 × weeks. **Slope** 2 means two predicted centimeters per additional week in this data range.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

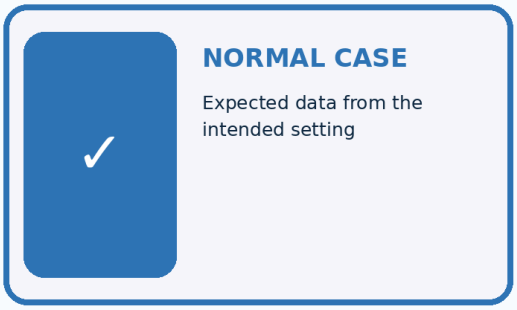
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

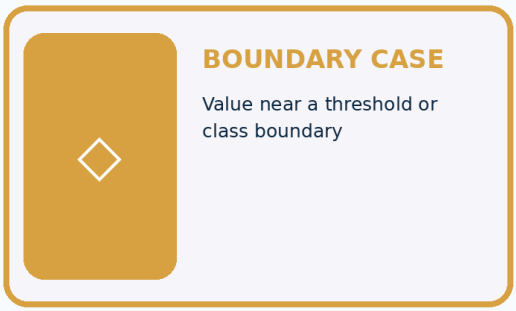
Compare selecting a line with tracing a supplied equation

> A zero training score does not establish a growth law.

Four evaluation conditions



NORMAL Expected data and expected behavior.

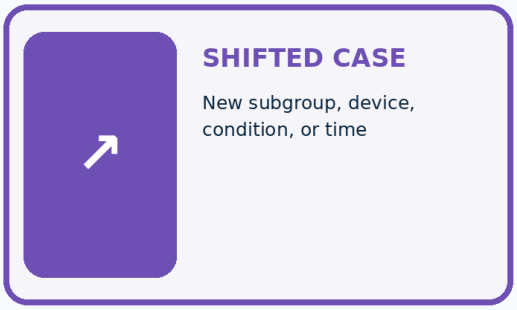


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves **generalization**. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Evaluation **observation** becomes8 instead of6.

▼ **Check after predicting**

Prediction: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

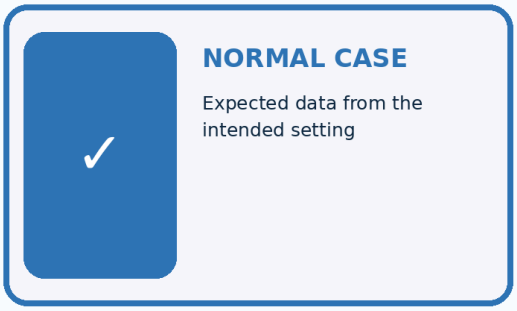
Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

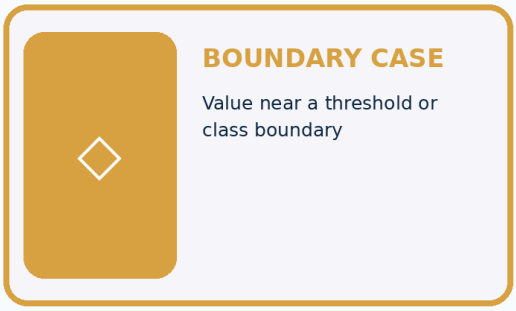
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

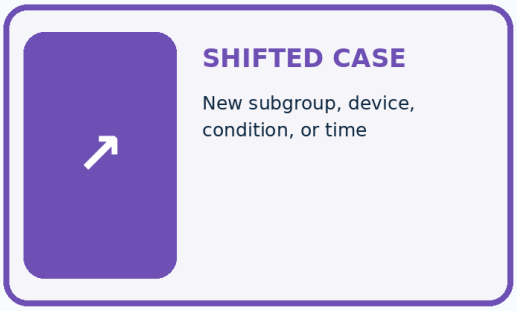
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace the same stored line before reading a changed outcome

> Evaluation targets change the error calculation, not the fitted state.

Complete pictured facts and main-fixture scope

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

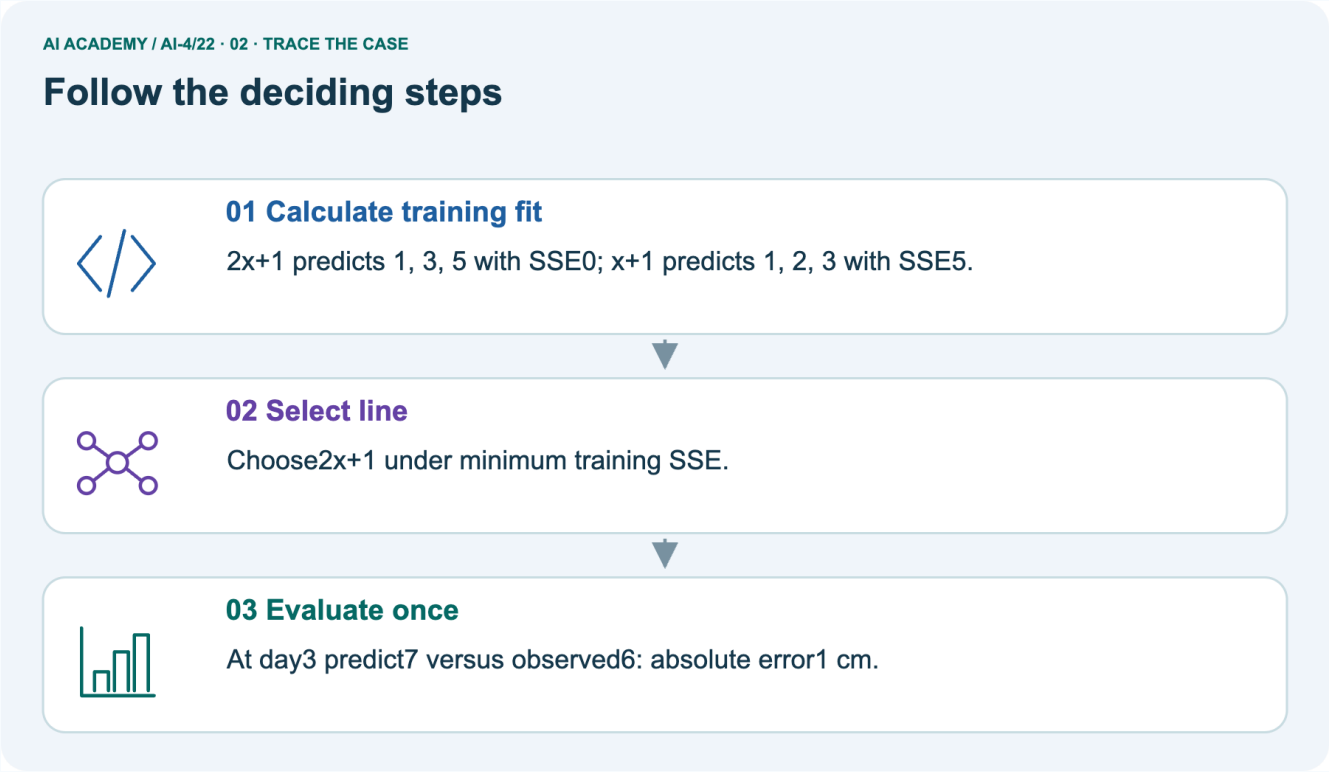
Training(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair:(3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main training [(0,1),(1,3),(2,5)] chooses (m,b)=(2,1); main validation [(3,6),(4,10)] gives predictions [7,9], residuals [-1,+1] cm, MAE1cm and MSE1cm². Training mean baseline3cm gives residuals [3,7], MAE5cm,MSE29cm². Picture uses only day3 observation6, later changed to8: prediction7 stays frozen, absolute error1cm in both cases, while residual changes-1→+1cm.

Main validation is a tiny extrapolation beyond training days0–2; it does not justify a biological growth law or long-range forecasts.

Trace candidate fit and day3prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Calculate training fit	2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5.
2. Select line	Choose2x+1 under minimum training SSE.
3. Evaluate once	At day3 predict7 versus observed6: absolute error1 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

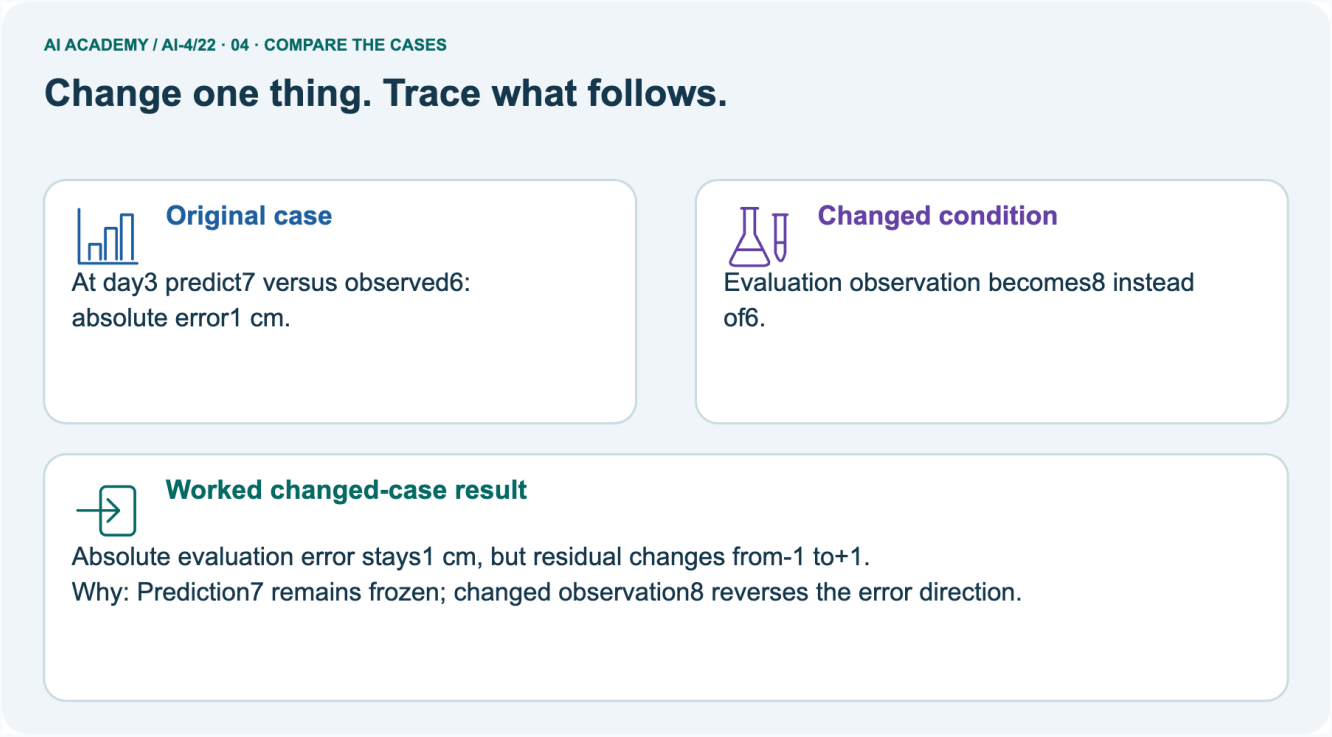
Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict changing only the day3observation

- Change one thing: Evaluation **observation** becomes8 instead of6. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State evaluation and extrapolation limits

- An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

- Read the labelled picture: May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the complete two-query validation question

Fitting a line at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Line A	Slope 1 intercept 1	Training SSE 5
Line B	Slope 2 intercept 0	Training SSE 3
Line C	Slope 2 intercept 1	Training SSE 0

For separate validation pairs (3, 6),(4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Predict at x 3 and x 4 and calculate MAE/MSE. Explain the steps or evidence behind your response.

▼ **Earlier wording, alternatives and picture transcripts**

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 22 • WORKED EVIDENCE

1. Model predicts height = 4 + 2 × weeks.
2. Slope 2 means two predicted centimeters per additional week in this data range.
3. At week 5, prediction is 14 cm; if actual is 16, residual is +2 cm.
4. Evaluate unseen plants and compare test MAE with the median baseline.

STEP 1 Model predicts height = 4 + 2 × weeks.

STEP 2 **Slope** 2 means two predicted centimeters per additional week in this data range.

STEP 3 At week 5, **prediction** is 14 cm; if actual is 16, residual is +2 cm.

STEP 4 Evaluate unseen plants and compare test MAE with the median baseline.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

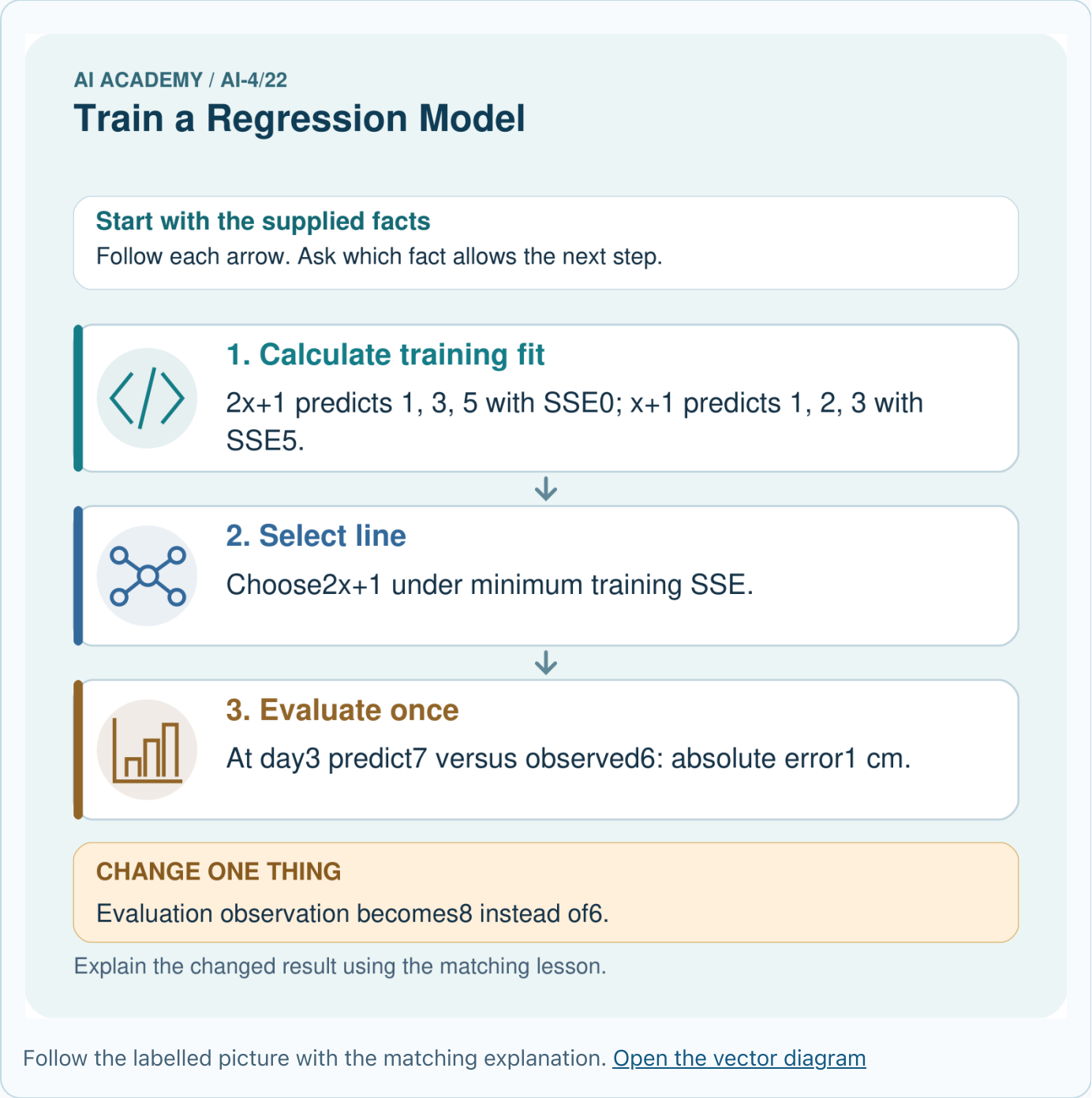
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train a Regression Model

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a fitted score mistaken for a general relationship

> The candidate set and intended-use range limit the claim.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE A positive **slope** proves one variable causes the other.

Explain evidence and next-check limits

- Record candidates, **training** rows, tie policy and chosen parameters. Do not add a candidate after seeing validation and report it as if only the original search happened.

  A zero **training** loss can coexist with validation mistakes.

> **Why is an exact training line not proof of correct predictions at x 100?**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: A positive slope proves one variable causes the other. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise validation and interpret coefficients with units

> A slope uses target units per feature unit; an intercept uses the target unit.

Try W2 with V1/V2observations beside you

Use workbook W2 for the connected example “**Fitting** a line”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Interpret days,cm and extrapolation

If x measures days and y measures cm, **slope** 2 means two predicted cm per day and **intercept** 1 means one predicted cm at day zero.

Predicting far outside the observed x range is extrapolation and may fail even when the **training** fit is exact.

Explain the fitting/evaluation evidence

- Use the worked case for **Fitting** a line. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain which labels may select the picture’s line

- May day3’s evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the new independent training/candidate set

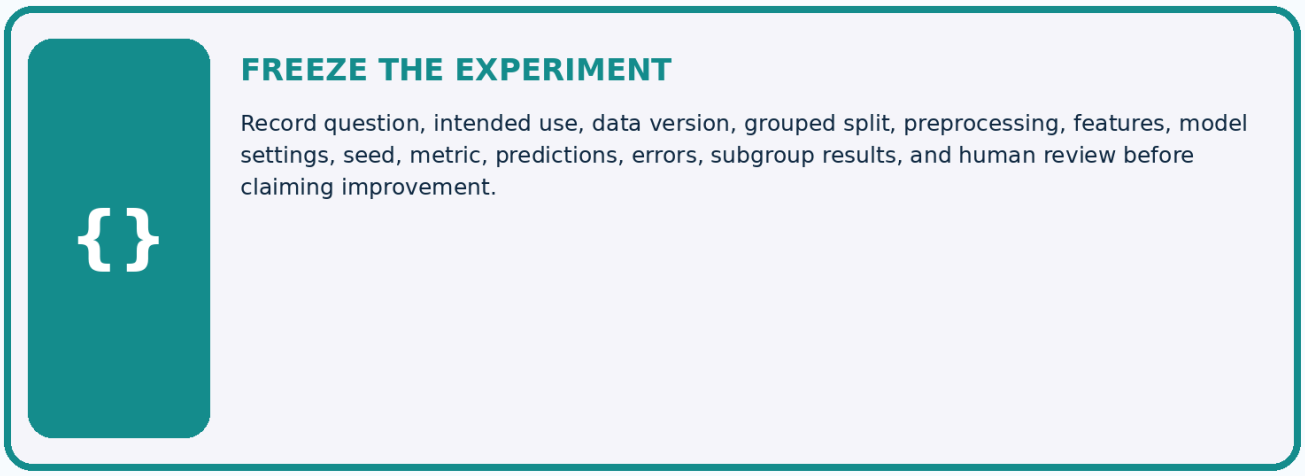
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Score validation

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run complete candidate fitting and same-case baseline scoring

> The source now supplies every target needed for the named investigation.

Complete fit,predict,evaluate and baseline record

Regression Run stage	Complete supplied values / checked evidence
Training and candidate contract	T1(0,1),T2(1,3),T3(2,5);A(1,1),B(2,0),C(2,1) as slope/intercept;score training SSE;first listed wins ties.
Training-only selection	Candidate SSE5,3,0cm² selects C:y=2x+1.
Frozen validation line	V1(3,6),V2(4,10)→predictions7,9cm;residuals-1,+1;MAE1cm,MSE1cm².
Training-derived mean baseline	Training y values1,3,5→constant3cm;validation predictions3,3;errors3,7;MAE5cm,MSE29cm².
Units and scope	Slope2cm/day,intercept1cm at day0. Candidate line search is not a full least-squares solver. No final test or broad biological evidence supplied.

Normal,tied,missing and far-query probes

Complete line-fitting probe	Expected bounded check
Normal training fixture	SSE5,3,0selects(2,1)before validation labels.
Candidate-score tie:training(0,1),candidates(1,1),(2,1)	Both SSE0;first listed wins under declared policy. No evidence from this point chooses a slope.
Missing empty training/candidate/evaluation collection	No fitted choice or observed error average;report unavailable/request data.
Far query x100days	Stored line predicts201cm algebraically;far outside training0–2days,no validity or biology claim.

Calculate,select,freeze and compare

1. Regression Run: fill each training prediction and squared error before selecting the candidate. Freeze its slope/intercept,then predict V1/V2without refitting. Score both the line and training-mean baseline on the same two observed targets. State units for slope,intercept,residual,MAE and SSE/MSE.

   The complete function version keeps validation out of the fit arguments;its numeric/unit contract still requires source checking. No global line optimization,causal growth model or untouched final-test result is claimed. Keep the independent m3/b2task for an unaided first attempt.

Optional complete fitting/evaluation program

```
from statistics import mean


def fit_training_line(training, candidates):
    if not training or not candidates:
        raise ValueError('Need nonempty training and declared candidate set')
    scored = []
    for m, b in candidates:
        sse = sum((y - (m * x + b)) ** 2 for x, y in training)
```

```
        scored.append(sse)
    best_index = min(range(len(candidates)), key=lambda i: scored[i])
    return candidates[best_index], scored


def evaluate(line, evaluation):
    if not evaluation:
        raise ValueError('No observed error average without evaluation records')
    m, b = line
    predictions = [m * x + b for x, y in evaluation]
    residuals = [y - p for (x, y), p in zip(evaluation, predictions)]
    return predictions, residuals, mean(abs(r) for r in residuals), mean(r*r for r in
residuals)


# Caller must supply finite numeric pairs in declared day/cm units.
training = [(0, 1), (1, 3), (2, 5)]
candidates = [(1, 1), (2, 0), (2, 1)]
line, training_sse = fit_training_line(training, candidates)
validation = [(3, 6), (4, 10)]
print(training_sse, line)
print(evaluate(line, validation))
constant = mean(y for x, y in training)
print('training mean', constant, evaluate((0, constant), validation))
```

Retain evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.
</details>

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Regression Run: trace code, interpret coefficients with units, predict unseen values, and compare MAE with baseline.

1. Write the hypothesis and expected **evidence** before running.

2. Regression Run: trace code, interpret **coefficient**s with units, predict unseen values, and compare MAE with baseline.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: select and evaluate a line on fresh pairs

> Show the candidate scores and frozen validation predictions.

Complete independent dataset and scope

Training (0, 2),(1, 5),(2, 8). Candidates m 2 b 2, m 3 b 1, m 3 b 2. Select by SSE, then score (3, 10),(4, 15) with MAE.

Use only the selected training(0,2),(1,5),(2,8),its listed slope/intercept candidates and validation(3,10),(4,15). Select by training SSE with earlier-candidate ties before scoring the new targets. These numbers differ from the main(2,1)line example. No exact biological law or broad performance claim follows from this small candidate search.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 22 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A regression model has validation MAE 2.4 cm; a **training**-mean baseline has validation MAE 2.0 cm on the same examples. Which performs better on this metric?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Add the regression experiment to the project appendix only as a separate, clearly limited model.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Training points (x,y)=(1,3),(2,5),(3,7). Compare y=x+1 and y=2x+1 by **training** MAE; select one, then predict held-out x=4, actual=10. Explain what was fitted.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep line selection and evaluation as separate evidence

> A reader should be able to recompute the chosen candidate and its held-out errors.

Project artifact and record

THIS WEEK'S CHECKPOINT Add the regression experiment to the project appendix only as a separate, clearly limited model.

Save the numeric experiment as a separate project appendix. Record target/time/units,training/evaluation sources,candidate list and order,training SSEs,tie policy,selected slope/intercept and each validation prediction/error. Keep the same-case constant baseline and extrapolation limits.

Later validation-informed changes require a new development record and a protected final evaluation,not a rewritten training-only history.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

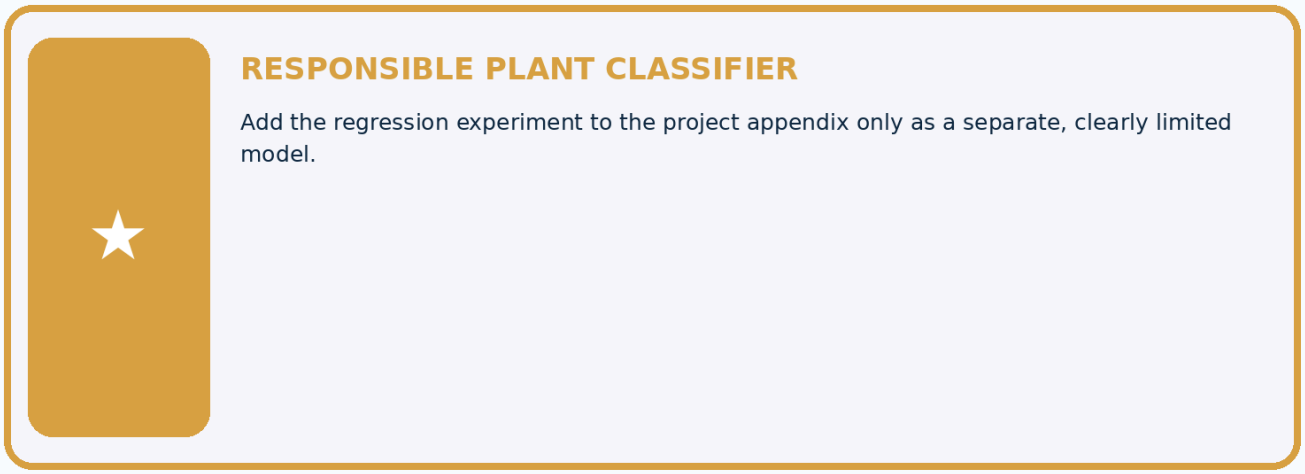
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

Finish by naming what was selected and what was evaluated

> Explain the candidate-set limit and the slope/intercept units.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Fit a line from a declared **training candidate set**

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
coefficient	learned relationship weight	
intercept	prediction when inputs are zero	
slope	predicted change per input unit	
generalization	performance on unseen cases	

Closer-explanation terms

Word	Meaning
Slope	Change in predicted target per unit change in x
Intercept	Predicted target when x is zero
Candidate set	The declared model choices searched

Word	Meaning
Fitting	Selecting model values using training evidence

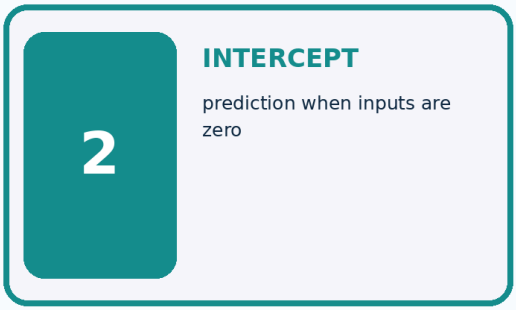
▼ Earlier wording, alternatives and picture transcripts

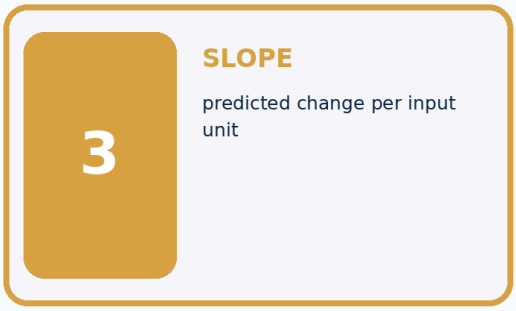
VOCABULARY

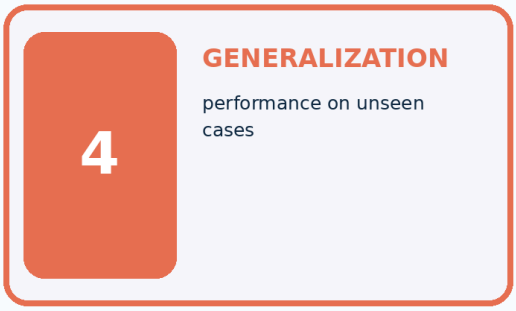
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR A positive **slope** proves one variable causes the other.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **coefficient**, **intercept**, **slope**, **generalization**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE A positive **slope** proves one variable causes the other.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

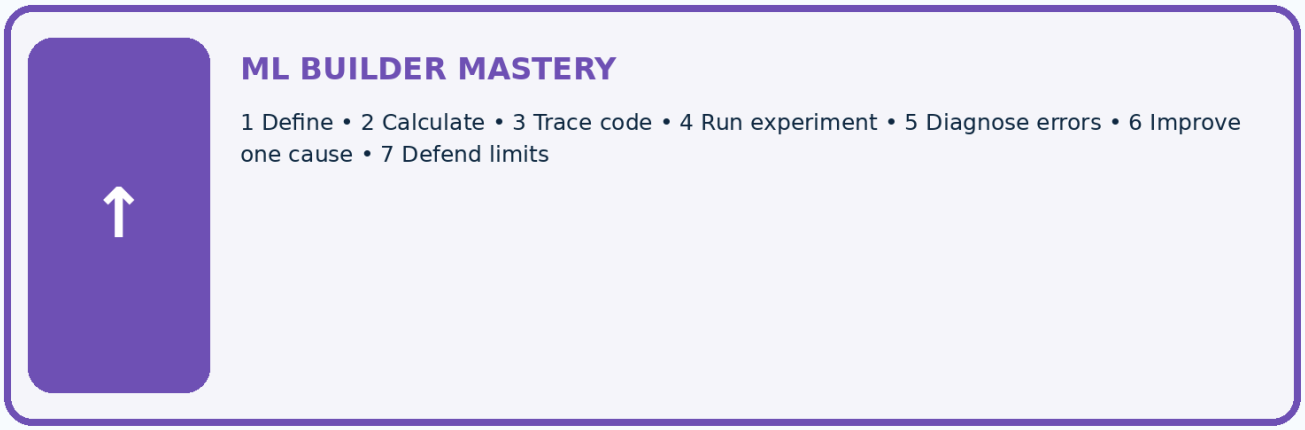
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh m3/b2line fitting and validation task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

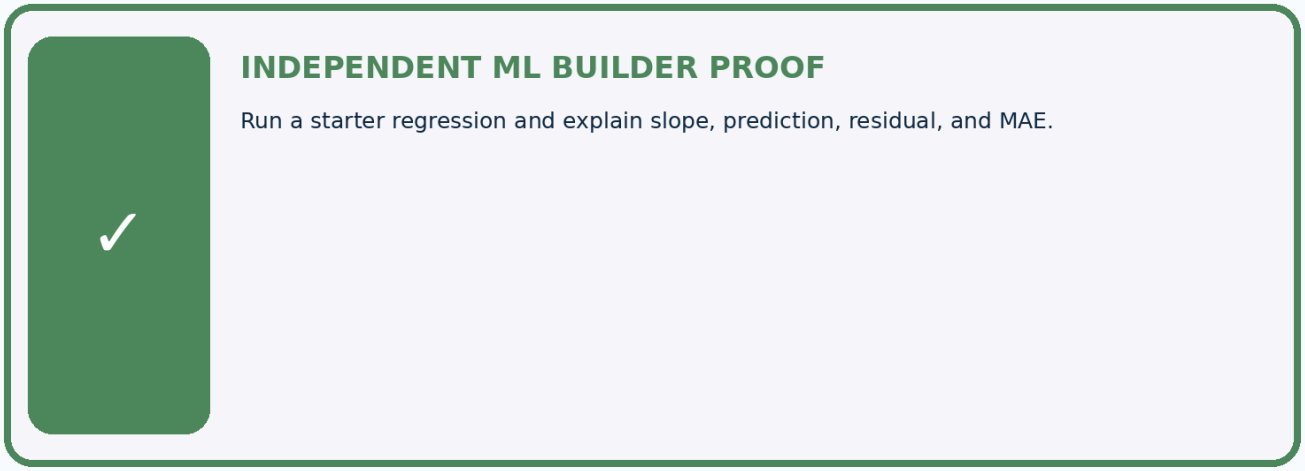
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Run a starter regression and explain **slope**, **prediction**, residual, and MAE.

NEXT BRIDGE Carry the experiment **evidence** into: Hold Out Test Data.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

Evidence or stage	Value or result
Rule	height = 2 × age + 1
Age 3	Predict 7, actual 8, absolute error 1
Age 4	Predict 9, actual 9, absolute error 0
MAE	0.5 cm

▼ Earlier wording, alternatives and picture transcripts

Week 22 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Apply the fitted equation, then evaluate

Supplied fitted rule: height=2 times age+1. Held-out **observation**s are (3,8),(4,9).

Predict and evaluate; distinguish inference.

Predictions 7,9; errors 1,0; MAE=.5 cm. Applying a supplied equation does not show how it was fitted.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 16

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Fit a line from a declared training candidate set	Explain the deciding step, show a trace or test, and state one limit.
Separate training loss from validation comparison	Explain the deciding step, show a trace or test, and state one limit.
Interpret slope and intercept with units	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 18

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**.

- Which labelled step matches this detail? At day3 predict7 versus observed6: absolute error1 cm.
- 1. Select line
- 2. Calculate **training** fit
- 3. Evaluate once
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Evaluate once At day3 predict7 versus observed6: absolute error1 cm. Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Evaluation **observation** becomes8 instead of6. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.
- 3. Absolute evaluation error stays1 cm, but residual changes from-1 to+1.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Absolute evaluation error stays1 cm, but residual changes from-1 to+1. **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Repair and keep the record

Review Absolute Error and MAE (ai-4/21). Goal: Calculate MAE MSE and RMSE. Short practice: Why is quoting MSE as4.67 cm misleading?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work
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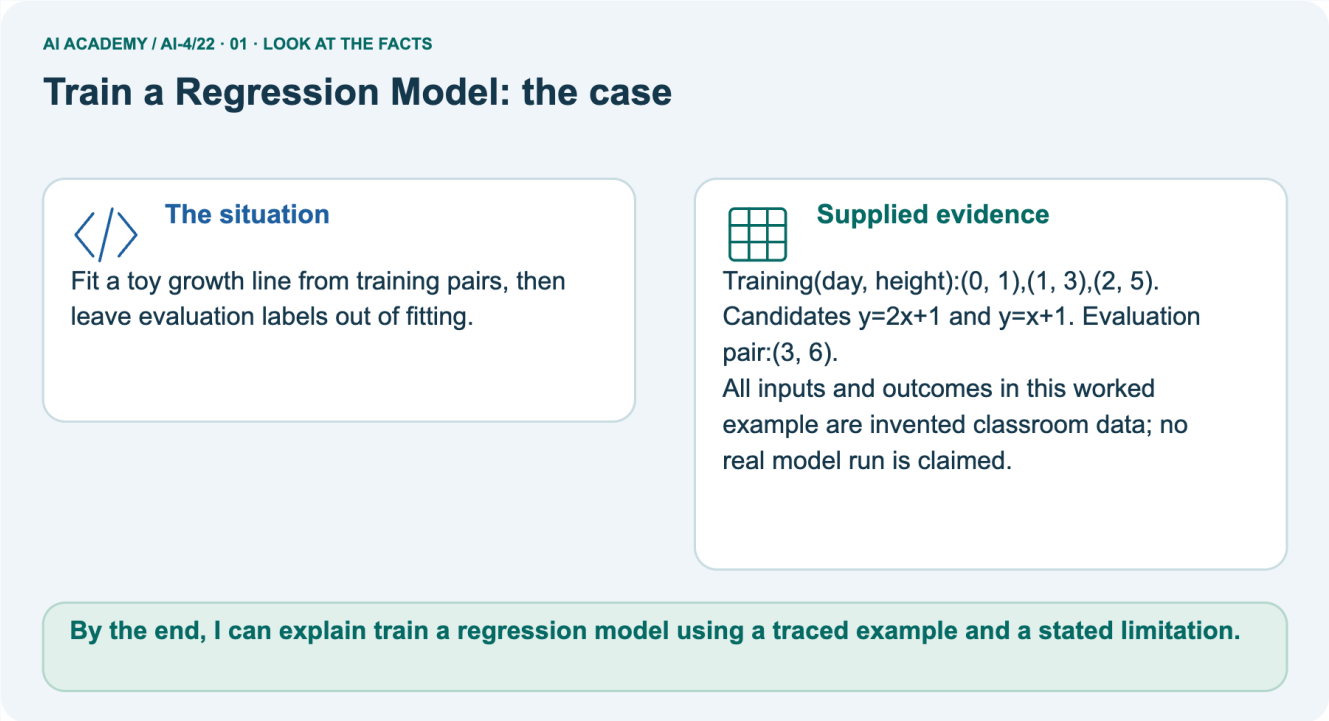
<details>
<summary>Reference · The six-picture journey</summary>
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REFERENCE · SECTION 19

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Train a Regression Model: the case



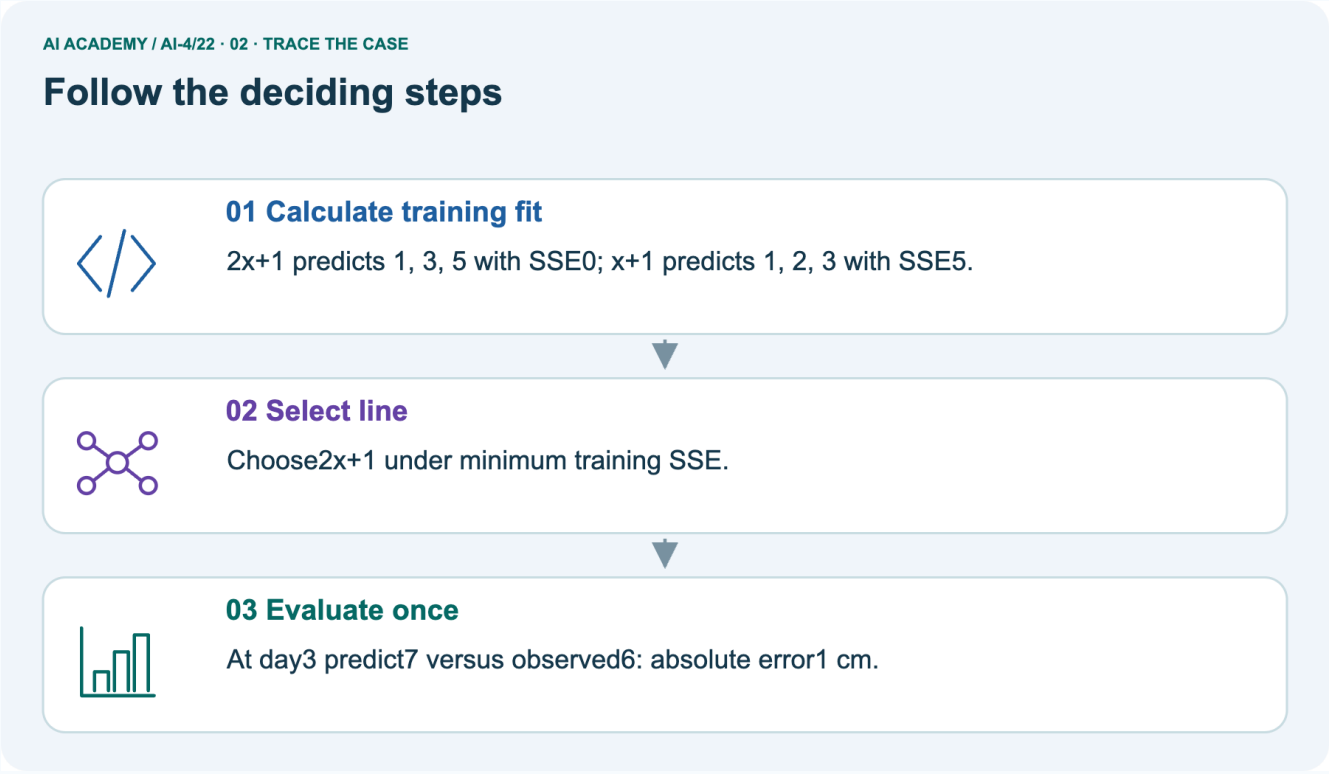
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit a toy growth line from **training** pairs, then leave evaluation labels out of **fitting**. Supplied facts: **Training**(day, height):(0, 1),(1, 3),(2, 5). Candidates y=2x+1 and y=x+1. Evaluation pair: (3, 6).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate **training** fit: 2x+1 predicts 1, 3, 5 with SSE0; x+1 predicts 1, 2, 3 with SSE5. → Select line: Choose2x+1 under minimum **training** SSE. → Evaluate once: At day3 predict7 versus observed6: absolute error1 cm.

Point to each numbered stage and explain which recorded **input** it uses.

03. Fitting a line at a glance

AI ACADEMY / AI-4/22 · 03 · INSPECT THE LESSON MODEL

Fitting a line at a glance

Part	Evidence or action	Interpretation
Line A	Slope 1 intercept 1	Training SSE 5
Line B	Slope 2 intercept 0	Training SSE 3
Line C	Slope 2 intercept 1	Training SSE 0

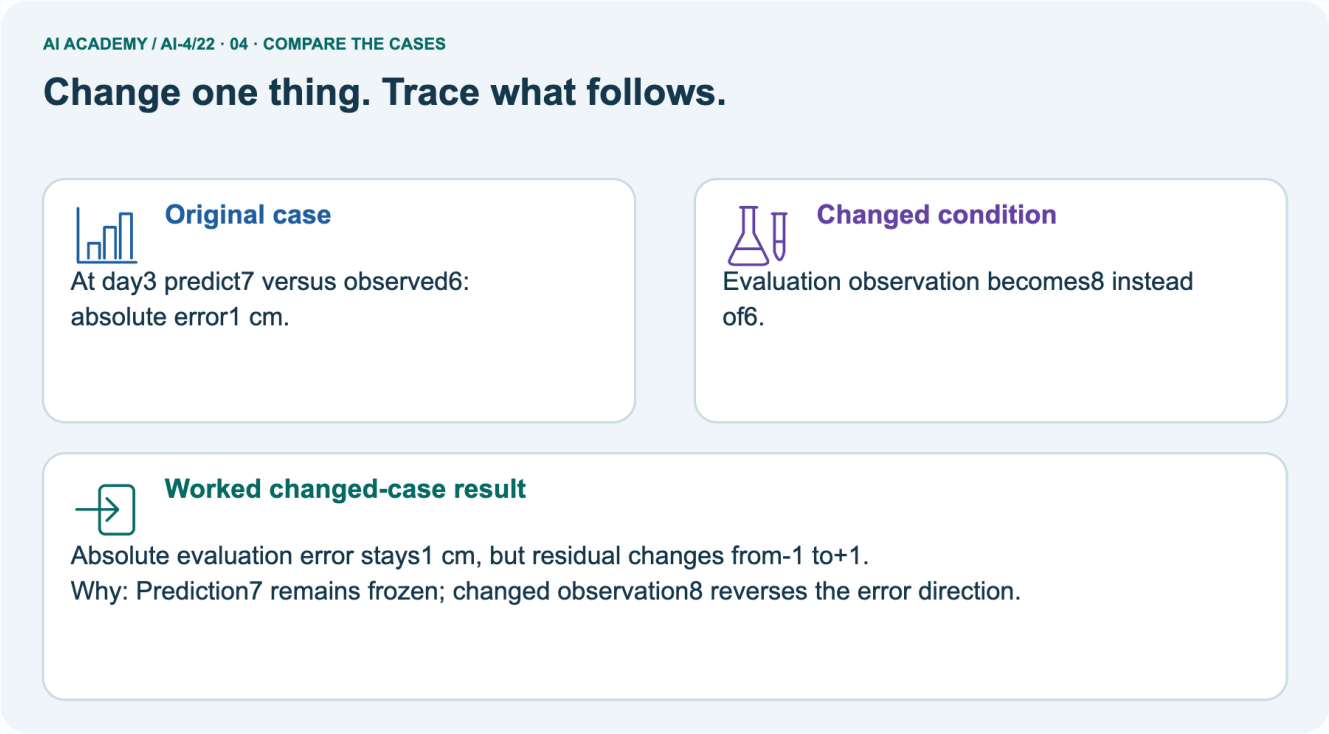
For separate validation pairs (3, 6),(4, 10), C predicts [7, 9], giving residuals [-1, 1], MAE 1 and MSE 1. The training-mean constant baseline is 3 and gives residuals [3, 7], MAE 5 and MSE 29. The fitted line wins on this small evaluation, without proving performance elsewhere.

Follow the labelled picture with the matching explanation. Open the vector diagram

Fitting a line at a glance. Line A | **Slope** 1 **intercept** 1 | **Training** SSE 5. Line B | **Slope** 2 **intercept** 0 | **Training** SSE 3. Line C | **Slope** 2 **intercept** 1 | **Training** SSE 0

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



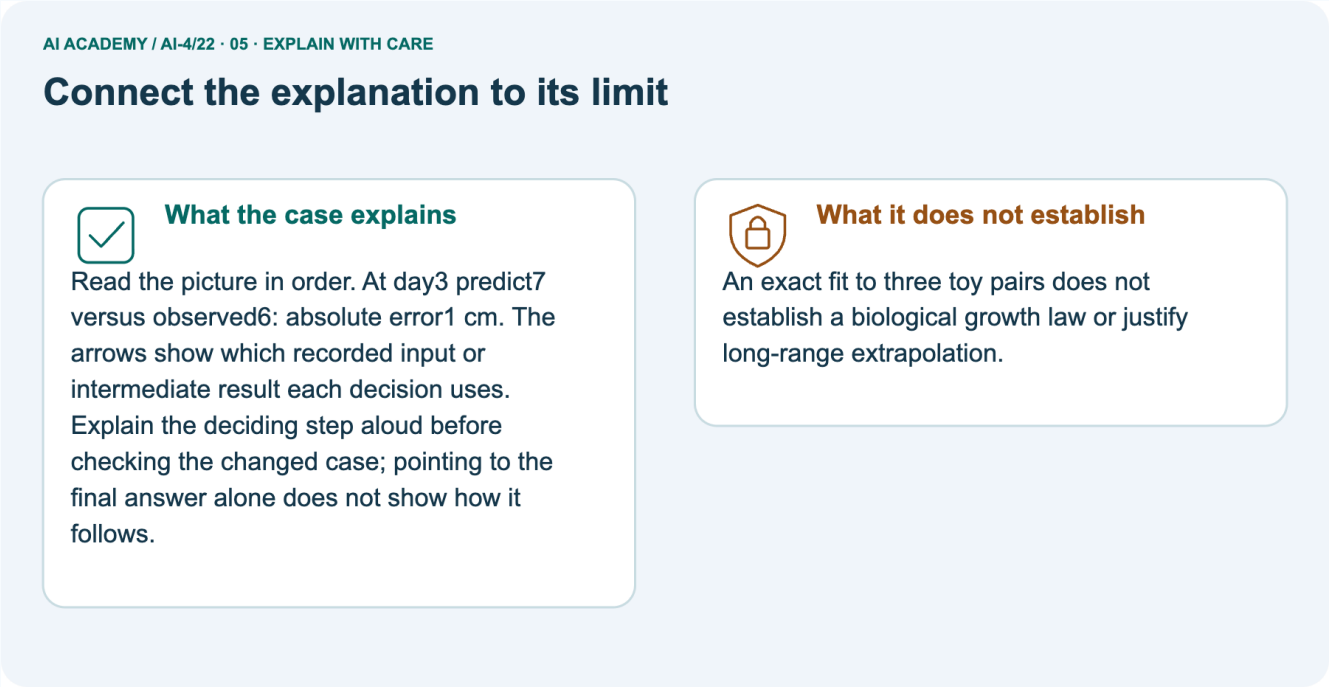
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At day3 predict7 versus observed6: absolute error1 cm. Changed condition: Evaluation **observation** becomes8 instead of6. Expected result: Absolute evaluation error stays1 cm, but residual changes from-1 to+1.

Reason: **Prediction**7 remains frozen; changed **observation**8 reverses the error direction.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

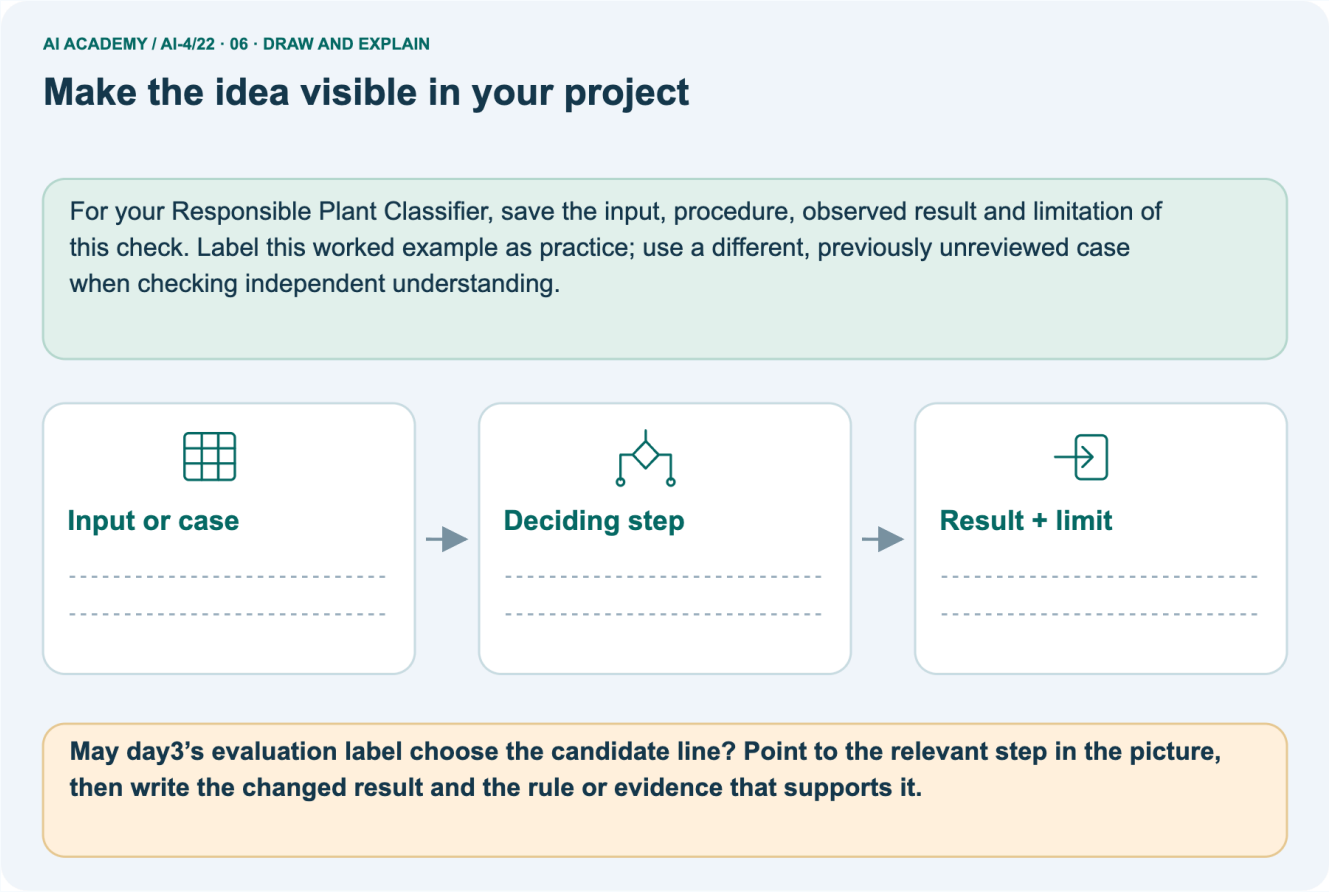
Read the picture in order. At day3 predict7 versus observed6: absolute error1 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: An exact fit to three toy pairs does not establish a biological growth law or justify long-range extrapolation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: May day3's evaluation label choose the candidate line? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Train a Regression Model: the case

02. Follow the deciding steps

03. Fitting a line at a glance

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Extension · Fit a line with the least-squares formula

REFERENCE · SECTION 15

Extension · Fit a line with the least-squares formula

> A complete retained math bridge shows a different fitting method and its own training/week units.

Read the method and complete arithmetic bridge

Estimate **slope** and **intercept** from **training** pairs using least squares. Ages are weeks and heights are cm. Test targets are excluded from **fitting**; **training** ages must include distinct values.

A straight line predicts y = **slope** × x + **intercept**. **Slope** is the change in predicted height for one extra week; **intercept** is the **prediction** at x=0, which may lie outside the measured ages.

You need means, subtraction, multiplication and squared numbers for this example; calculus is not required.

x (weeks) · y (cm) · x – mean x · y – mean y · Product · Squared x difference

> **x (weeks):** 1
>
> **y (cm):** 3
>
> **x – mean x:** –1
>
> **y – mean y:** –2
>
> **Product:** 2
>
> **Squared x difference:** 1

> **x (weeks):** 2
>
> **y (cm):** 5
>
> **x – mean x:** 0
>
> **y – mean y:** 0
>
> **Product:** 0
>
> **Squared x difference:** 0

> **x (weeks):** 3
>
> **y (cm):** 7
>
> **x – mean x:** 1
>
> **y – mean y:** 2
>
> **Product:** 2
>
> **Squared x difference:** 1

Mean x=2 and mean y=5. Add the product column: 4. Add squared x differences: 2. **Slope**=4/2=2 cm per week. **Intercept**=5−2×2=1 cm.

This least-squares formula chooses the line minimising squared **training** errors when the x values are not all identical. Compare its held-out error with a baseline; a good **training** fit alone is insufficient.

This separate extension estimates slope/intercept with the least-squares formula rather than searching the main three candidates. Its age unit is weeks, while the main classroom fixture uses days. Keep the records and coefficient units separate. The supplied one-case test is a teaching fixture whose label is visible, not a protected real final evaluation.

Complete formula-based fit/test script

```
"""Fit a line to training pairs and evaluate a separate example."""
train = [(1, 3), (2, 5), (3, 7)]
if not train:
    raise ValueError("Training examples are required")
n = len(train)
x_mean = sum(x for x, y in train) / n
y_mean = sum(y for x, y in train) / n
num = sum((x - x_mean) * (y - y_mean) for x, y in train)
den = sum((x - x_mean) ** 2 for x, y in train)
if den == 0:
    raise ValueError("Distinct training ages are required")
slope = num / den
intercept = y_mean - slope * x_mean
test = [(4, 10)]
if not test:
    raise ValueError("Test examples are required for MAE")
preds = [slope * x + intercept for x, y in test]
mae = sum(abs(y - pred) for pred, (x, y) in zip(preds, test)) / len(test)
print(slope, intercept, preds, mae)
```

Trace output and its one-case limit

Expected **output**: 2.0 1.0 [9.0] 1.0

The fitted line is height = 2 times age + 1. At four weeks it predicts 9 cm; the observed height of 10 cm gives an absolute error and one-case MAE of 1 cm. **Fitting** a line and applying a supplied line are different activities.

One held-out case is very limited **evidence**.

Independent code-change question and response

- Change the held-out height from 10 to 8 without changing the **training** data. Predict the fit, **prediction** and MAE.

- My **prediction** and **evidence**

Write or draw your reasoning and evidence.

Earlier wording, alternatives and picture transcripts

Fit and test a straight-line regression model

Math bridge: fitting the line

Independent code change

Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.